

Course Title	VLSI Design
Credit Structure	3-0-0-3
Course Code	ED312
Program	B.Tech (EVD)
Category	Core

Course Objective: This course introduces design fundamentals necessary to implement VLSI circuits.

Course Contents:

CMOS Circuits: Timing issues, pipelining, resource sharing, metastability, synchronization,

Semi & full custom designs: Placement & routing, Algorithms for Physical Design,

IO pads & ESD

Static & Dynamic CMOS Transistor, Ratioed circuits, domino logic, pass transistor, latches

Memory Designs: SRAM, DRAM, ROM, PROM, EPROM, EEPROM, Flash, CAM, VLSI Layout of small digital systems using predefined cells using a layout editor, System Verilog, CAD Tools, HDL

Textbooks

1. Wakerly J. F ~ Digital Design: Principles and Practices (4th Edition, Pearson, 2008)
2. Neil H.E. Weste & David Money Harris ~ CMOS VLSI Design: A Circuits and Systems Perspective (Addison Wesley, 4th Edition, 2011)

Evaluation Components

In semester evaluations – 60%

End semester evaluation – 40%

Program Outcomes

- Analyze the operation and timing of CMOS circuits, including metastability, synchronization, pipelining, and resource sharing in synchronous digital systems.
- Design and evaluate static and dynamic logic circuits such as CMOS, ratioed logic, domino logic, and pass-transistor logic, including flip-flops and latches.
- Develop memory architectures with a focus on their layout, read/write mechanisms, and integration.
- Apply algorithms for placement and routing in physical design.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X		X		X	X						X